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I&D (SSIS + SSAS + SSRS + PowerBI) LoT Course Structure

I&D (SSIS + SSAS + SSRS + PowerBI) LoT provides exposure to a band of data warehousing technologies. It focuses on application development for data warehouses. The following table lists the course structure for I&D LoT.

Sr. No.	Course	Duration (In Days)	Remarks
1	Discover(Induction)	-	Online
2	Soft Skills Part 1	1	Soft Skills Part 1
3	Introduction to Python	2	
4	SQL server 2016 for BI	10	Project kick off
5	Module 1 Test	1	Coding and MCQ
6	Soft Skills Part 2	1	Soft Skills Part 2
7	Data Warehouse Concepts	1	Sprint 1 Implementation, MCQ
8	Data Modelling for Business Intelligence	1.5	
9	ETL Basics	0.25	
10	Software Testing for BI	0.25	
11	Introducing to cloud	1.5	
12	Introduction to Software Engineering	0.5	
13	Soft Skills Part 3	1	Soft Skills Part 3
14	SSIS	8	Sprint 1 Implementation, MCQ
15	Sprint 1 Evaluation	2	Sprint 1 Evaluation
16	Soft Skills Part 4	1	Soft Skills Part 4 (Evaluation)
17	SSAS	4	Sprint 2 Implementation, MCQ
18	SSRS Introduction	2	
19	PowerBI	2	
20	Sprint 2 Evaluation	2	Sprint 2 Evaluation
21	SSRS to PowerBI Migration Overview	1	
22	Big Data Overview	2	
23	SSIS to ADF migration Overview	1	
23	L1 Preparation	1	
24	L1 Test	1	MCQ
Total Training Duration		**48**	

I&D Curriculum

Introduction to Python

Program Duration: 2 days.
Contents:

Introduction to Python Programming
- Why do we need Python?
- Program structure in Python

Execution steps
- Interactive Shell
- Executable or script files.
- User Interface or IDE

Data Types and Operations
- Numbers
- Strings
- List
- Tuple
- Dictionary
- Other Core Types

Statements and Syntax in Python
- Assignments, Expressions and prints
- If tests and Syntax Rules
- While and For Loops
- Iterations and Comprehensions

Functions in Python
- Function definition and call
- Function Scope
- Arguments
- Function Objects
- Anonymous Functions

Modules and Packages-Basic
- Module Creations and Usage
- Package Creation and Importing

Classes in Python
- Classes and instances
- Classes method calls

File Operations
- Opening a file
- Using Files

· Other File tools

Libraries

- Importing a library
- Math
- Numpy

SQL server 2016 for BI

Program Duration: 10 days.
Contents:

Lesson 01: Introduction to RDBMS

- Introduction to databases
- Data Models in Database
- Properties of RDBMS
- Normalization
- CODD's Relational Database Rules
- Data Integrity
- Transact-SQL (T-SQL)

Lesson 02: Introduction to E-R Modeling, Normalization (up to 3 NF)
Lesson 03: Introduction to SQL Server 2016

- What is SQL Server?
- SQL Server Components
- SQL Server Authentication Modes
- SQL Server Services
- SQL Server Version History
- SQL Server Key Business Scenarios
- SQL Server Editions
- SQL Server Tool & Utilities
- SQL Server System Databases
- SQL Server – Database Objects
- SQL Server system Tables
- SQL Server - Metadata Retrieval
- SQL Server – Features

Lesson 04: Working with Data Types, Tables & Data Integrity covering DDL, DML, DCL statements.

- Working with Data Types
- Working with Schema
- Working with Tables
- Implementing Data Integrity

Lesson 05: Beginning with Transact-SQL

- Transact-SQL
- System Functions
- Advanced T-SQL Queries
- Advanced T-SQL Statements
- Other T-SQL Statements

Lesson 06: Working with Joins and Subqueries

- What are Joins?
- Types of joins
- Subqueries

Lesson 07: Set operators

- UNION, UNION ALL, EXCEPT, INTERSECT

Lesson 08: Database Objects: Indexes, Views and Sequences

- Introduction to Index in SQL Server
- Introduction to Views in SQL Server
- Working with Sequence

Lesson 09: Working with Batches and Cursors

- T-SQL statement in Batch
- Overview of Cursor

Lesson 10: Procedures and Functions

- Stored Procedure
- Implementing Stored Procedure
- Exception handling using TRY-CATCH
- Implementing User Defined Functions

Lesson 11: Transactions

- Managing Transactions
- TCL statements
- @@trancount global variable

Lesson 12: Implementing Triggers

- Introduction to Triggers
- Defining Triggers
- How Triggers Work?

SQL Server Profiler

- Where is the Profile
- Use SQL Server Profiler
- SQL Server Profiler concepts

SQL Server Optimization

Introduction- How the Query Optimizer Works

Execution Plans

Cost Estimation

Data Warehouse Concepts

Program Duration: 1 day.

Contents:

Business Intelligence
- Business Intelligence
- Need for Business Intelligence
- Terms used in BI
- Components of BI

General concept of Data Warehouse
- Data Warehouse
- History of Data Warehousing
- Need for Data Warehouse
- Data Warehouse Architecture
- Data Mining Works with DWH
- Features of Data warehouse
- Data Mart
- Application Areas

Dimensional modeling
- Dimension modeling
- Fact and Dimension tables
- Database schema
- Schema Design for Modeling
- Star
- Snow Flake
- Fact Constellation schema

ETL and Metadata
- ETL process
- Metadata used in ETL
- Metadata in Data Warehousing
- Simple Data warehouse model

Online Analytical Processing (OLAP)
- Online Analytical Processing (OLAP)
- Nature of OLAP analysis
- Types of OLAP
- OLAP Tools
- OLTP and OLAP
- OLAP Functional requirements
- OLAP Fast and Selective
- Operational versus Informational System

Data Mining
- Data mining

The Knowledge Discovery process
Need of Data Mining
Use of Data mining
Data mining and Business Intelligence
Types of data used in Data mining
Data Mining applications
Data Mining products
Data Mining market

Best Practices for Building Data Warehouse

Recipe for a Successful data warehouse
Data warehouse pitfalls
Popular BI DW tools and suits
Trends in BIDW

Data Modeling for Business Intelligence

Program Duration: 1.5 days

Contents:

- Introduction to Data Modeling
  - Importance of data modeling
  - Features of a good data model
  - Who should be involved in data modeling
  - Database design stages and deliverables
  - Classification of information
- Understanding Business Requirements
  - Need of Requirement Analysis
  - Characteristics of a Good Requirement
  - The Data Life cycle
  - Methods of Collecting requirement
  - Business Requirement Specification (BRS)
- Conceptual Model
  - Define conceptual model
  - Objectives of conceptual model
  - Components of Conceptual Model
  - Types of Modeling
  - Entity-Relationship (ER) model
  - Types of Attributes
  - Join Problems
  - Steps of dimension modeling

Star Schema
Snowflake Schema
Bill Inmon Vs Ralph Kimball Approach

- Logical Model
  - Define logical model
  - List features of a logical model
  - Transformations required to be done while converting a conceptual model into a Logical model
  - Activities in table specification
  - Activities in column specification
  - Activities in Primary key specification

ETL Basics

Program Duration: 0.25 day
Contents:

- Basic Concepts
  - Data warehouse
  - Data warehousing strategies
  - Data warehouse architecture
  - ETL Meaning
  - Need for ETL
  - ETL Process
  - Operational Considerations
- ETL Process
  - Data extraction
  - Data transformation
  - Data Loading
- Operational Considerations
  - Exceptional Handling
  - Alerts and Notification
  - Process restart-ability
  - Job Scheduling and Monitoring

ETL Tools
- Leading ETL tool vendors
- ETL tool strengths / weaknesses
- Choosing the correct ETL tool

Software Testing for BI

Program Duration: 0.25 day
Contents:

- Introduction to Software testing for BI
  - Business requirements
  - BI Project versus BI Program
  - How is BI testing different from traditional code based testing?
  - BI SDLC
- Testing concepts
  - What is testing? Testing – Why? Testing – How?
  - Principles of Testing
  - Test Case and Test Suite
  - Testing scope
  - Test Strategy
  - Verification and Validation
- Types of Testing
  - Static Testing, Dynamic Testing, Automated testing
  - V Model for BI Testing
- Testing for BI
  - Testing document purpose (Test documentation)
  - General BI Testing Principles
  - BI Testing Mission
  - Production Verification Testing
  - Possible Areas of Automation

Introduction to Cloud

Program Duration: 1.5 days

- Introduction
- The Need for Cloud Computing
- The Definition of Cloud Computing
- Evolution of Cloud and Cloud Models
- What is SAAS , PAAS and IAAS
- Who is responsible for Data in cloud?
- The Five Key characterists of Cloud Computing
- Benefits of Cloud Computing
- CapEx Vs OpEx
- Cloud DataCenters : Where are they?

- What is Gartner saying about Cloud Providers?

Introduction to Software Engineering

Program Duration: 0.5 day

Contents:

- To Understand the following :
  - What is Software Engineering (SE)
  - Common life cycle models
  - Phases in SE
  - Familiarizing Requirements Phase
  - Familiarizing Design Phase
  - Familiarizing Construction Phase
  - Familiarizing Testing and acceptance Phase
  - Review and Configuration Management Process

SSIS

Program Duration: 8 days

Contents:

Introduction to ETL and Integration Services

- Overview of BI
- ETL Concept
- Different ETL Tool providers
- Defining SQL Server Integration Services (SSIS)

SSIS background

- DTS Limitation
- Difference between DTS , 2005, 2008 R2 ,2012,2014 and 2016

Typical Uses of Integration Services

- Merging Data from Heterogeneous Data Stores
- Populating Data Warehouses and Data Marts Cleaning and Standardizing Data
- Automating Administrative Functions and Data Loading

Tools

- Business Intelligence Development Studio and SQL Server data Tool
- SQL Server Management Studio
- Import and Export Wizard
- Package Configuration Wizard
- Package Deployment Wizard

SSIS Objects

- Package
- Control Flow
- Data Flow
- connection Manager

Data Flow Task

- Data Flow Task

Data Flow Task -Source

- ADO NET Source
- Excel Source
- Flat File Source
- OLE DB Source
- Raw File Source
- Script Component
- XML Source

Data Flow Task - Transformation

- Aggregate
- Audit
- Cache Transform
- Character Map
- Conditional Split
- Copy Column
- Data Conversion
- Data Mining Query
- Derived Column
- Export Column
- Fuzzy Grouping
- Fuzzy Lookup
- Import Column
- Lookup
- Merge
- Merge Join
- Multicast
- OLE DB Command
- Percentage Sampling
- Pivot
- Row Count

- Row Sampling
- Script Component
- Slowly Changing Dimension
- Sort
- Term Extraction
- Term Lookup
- Union All
- Unpivot

Data Flow Task-Destination

- ADO NET Destination
- Data Mining Model Training Destination
- DataReader Destination
- Dimension Processing Destination
- Excel Destination
- Flat File Destination
- OLE DB Destination
- Partition Processing Destination
- Raw File Destination
- Recordset Destination
- Script Component
- SQL Server Compact Edition Destination
- SQL Server Destination

Operating system-level tasks

- File System Task
- FTP Task
- Send Mail Task

SQL Server Tasks

- Bulk Insert Task
- Execute SQL Task
- Transfer Database Task
- Transfer Error Messages Task
- Transfer Jobs Task
- Transfer Logins Task
- Transfer Master Stored Procedures Task
- Transfer SQL Server Objects Task

Scripting Tasks

- Script Task

Profiling Task

- Data Profiling Task

Workflow Tasks

- Execute Package Task

- Execute Process Task
- Message Queue Task

Processing XML

- XML Task

Analysis Services Tasks

- Analysis Services Processing Task
- Analysis Services Execute DDL Task
- Data Mining Query Task

Maintenance Tasks

- Back Up Database Task
- Check Database Integrity Task
- Execute SQL Server Agent Job Task
- Execute T-SQL Statement Task
- History Cleanup Task
- Notify Operator Task
- Rebuild Index Task
- Reorganize Index Task
- Shrink Database Task
- Update Statistics Task

Precedence Constraint

- Configuring the Precedence Constraint

Variables and Expressions

- System Variables
- User Defined Variables
- Configuring Variables

SSIS Architecture

- SSIS Architecture overview
- Distinguishing between data flow pipeline and package runtime
- Executing packages on the client side or hosted in the SSIS service

Error Handling, Logging and Transactions

- Event Handler
- SSIS Logging
- transcation support and check point

Execution , debugging and Protection

- Package execution
- data Viewer
- Breakpoints
- Package Protection

Deployment

- Switching between project and package deployment modes

	o Deploying packages to the SSISDB o Running packages from SQL Server o Leveraging package parameters
Performance Tuning	o Designing SSIS Packages for High Performance o Monitoring the SSIS Performance with Performance Counters o Parallelize Destination Operations o Configure Rows per Batch and Maximum Insert Commit Size in OLEDB destination

SSAS

Program Duration: ~~4~~ 4 Days
Content

Introduction to SSAS	• What is BI? • The Microsoft BI Solution Stack • What is SSAS? • Components of SSAS
Designing a SQL Server Analysis Services Database	• Components of Designing a Analysis Services Database • Concept of Dimension • Attributes and Hierarchy • Concept of Cube • Measures and Measure Groups • Dimension Tables • Fact Tables • Star Schema and Snowflake Schema
Defining OLAP Dimension	• Defining Attribute Relationships • Special Dimension Types • Slowly Changing Dimensions
Using MDX Queries	• What is MDX Query? • MDX Query Terms: • Member • Tuple • Set • MDX Query Examples • Defining Calculated Members in SSAS Cube • Defining KPI in SSAS Cube
SSAS Cube Partitions and Storage Modes	• Managing Partitions for the Cube • Storage Modes for the Cube Partitions

Introduction to Data Mining using SSAS	• What is Data Mining? • Broad Steps involved in Data Mining • Understand: • Mining Structure • Mining Model • Mining Algorithms
Tabular Model Cube	• What is Tabular Cube • Components of Tabular Cube and different Modes of processing • Difference between SSAS and Tabular Cube • Introduction to DAX Query • Tabular Model Fact and Dimension Tables and Relationship • Cube Creation in Tabular Model • How to handle Role Playing Dimension in Tabular Model • Tabular Model Roles and Security • Tabular cube Partitioning • Tabular Cube Access via PowerPivot •

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SSRS

Program Duration: ~~4~~ 2 Days
Content

SSRS Introduction

i. Overview
   1. Report Manager
   2. Report Designer
   3. Browser Types
   4, Report Server
   5. Data Sources
   6. Report Server Database
   7. Command Line Utilities
ii. Setting up your environment
iii. My First SSRS report (Wizard Based)
iv. My First SSRS report (Non-Wizard Based)
v. Adding Parameters to SSRS reports
vi. Understanding SSRS cascading and multiple Parameters

vii. SSRS expressions
viii. Design of the SSRS Reports:
    1. Tables, Matrices and Lists
    2. SSRS Graphs and Visualization

3. SSRS Sub reports
4. Sub Report
5. Drill down Report
6. Drill through Report
7. Linked Reports
8. Ad-hoc Reports
9. Parameterized Reports

PowerBI

Program Duration: 2 days
Contents:

Power BI Overview
    Introduction to Power BI
    Power BI workflow
    Building Blocks of Power BI
    Power BI Architecture
    Product Portfolio
Power BI Components
    Power BI Components
    Power BI Desktop
    Data Model
    Data Model Viewer, Data Window
    Introduction to UI Objects
Data Connectivity
    Connect to Data Sources in Power BI
Power BI Features
    Quick Insights
    Data Exploration
    Sharing Dashboards
Easy Implementation Power BI and excel
    Publishing to cloud services
    Workbooks and Sheets
Road Map
    Pricing
    Power BI On Premise
Getting Started
    Creating Workbooks

- Comments in Power BI

Building Data Transformations

- Modelling
- Manage Data relationships
- Optimize data models
- Calculated Tables
- Time Based Functions
- Time Based Data Exploration

Creating Dashboards & Reports

- Reports
- Creating Dashboards
- Managing DashBoards , Sharing

Working with Various Charts & Custom Visualization

- Different Charts
- Pulling custom Visuals
- Managing Data in Charts
- Role Based Security

IoT and Analytics

- PowerBI with IoT /Azure
- R Visuals

PowerBI Embeded

Demo on POwerBI Embeded

SSRS to PowerBI Migration

Program Duration: 1 Day

Content

SSRS to PowerBI Migration Overview

Introduction to Paginated Reports in SSRS

1. Migration Approaches
    a. Automated Approach
    b. Manual Approach
2. Prerequisites for Migration
    a. PowerBI Report builder
    b. Configuration settings for enabling Paginated Reports workload
    c. Power BI Gateway to allow connectivity
3. Limitations of paginated reports

Big Data Overview

Program Duration: 2 days
Contents:

Big Data overview
Hadoop insight
Pig Latin
Hive query language
MapReduce
Introduction to Real-time Processing and brief on Kafka
Introduction to In-memory Processing
Introduction to Big data Analytics
One Big Data use case briefing

SSIS to ADF Migration Overview

Program Duration: 1 day
Contents:

1. Overview
    a. Azure SSIS lift-and-shift overview
2. Assessment
    b. Data Migration Assistant (DMA)
    c. Four storage types for SSIS packages
3. Migration
    d. Azure SQL Managed Instance as database workload destination
    e. Azure SQL Database as database workload destination